# 245127: porphyritic metadacite, Ellen Dam

## *(Black Flag Group, Kalgoorlie Terrane, Eastern Goldfields Superterrane, Yilgarn Craton)*

## Location and sampling

Kalgoorlie (SH 51-9), Kalgoorlie (3136)
MGA Zone 51, 351080E 6570361N

Sampled on 24 September 2020

This sample was collected from the 189.3 – 191.1 m depth interval of drillcore NGDD19021, drilled in 2019 by Northern Star Resources Limited. The drillhole is located on Woolibar Station, about 7.6 km east-northeast of Karramindie Soak, 3.0 km north of No. 2 Dam, and 2.2 km southeast of Ellen Dam.

## Geological context

The unit sampled is a metadacite assigned to the Black Flag Group (Morris, 1998; Hall et al., 2021). The 2692–2655 Ma Black Flag Group of the southwestern Eastern Goldfields Superterrane is an extensive northwest-trending unit, consisting of felsic metavolcaniclastic and metasedimentary rocks with subordinate felsic and mafic metavolcanic rocks (Squire et al., 2010). The Black Flag Group overlies the 2753–2680 Ma Kalgoorlie Group, is unconformably overlain by the 2664–2655 Ma Merougil Formation (Hall et al., 2021), and in part is intruded by the c. 2680 Ma Golden Mile Dolerite (Rasmussen et al., 2009). The sampled metadacite also includes a volcaniclastic component, indicated by the rare presence of cm-scale rounded pebbles of volcanic or volcaniclastic rock (possibly cognate inclusions). A porphyritic andesite of the Black Flag Group, collected about 43 km to the north-northwest, yielded an igneous crystallization age of 2690 ± 7 Ma (GSWA 211051, Lu et al., 2017).

## Petrographic description

The sample is a fine- to medium-grained, plagioclase-phyric metadacite (Fig. 1), consisting of about 40–50% groundmass, 20–25% carbonate minerals, 10–15% biotite, 5–7% actinolite, 5% plagioclase phenocrysts, 3% sericite, 2% sulfide minerals, 1–2% chlorite, 1–2% muscovite, and <1% magnetite. The groundmass appears to be quartz-rich, and may be silicified, but is obscured by secondary minerals, including carbonate, biotite, actinolite, sericite, and minor muscovite and chlorite. The groundmass is mainly microcrystalline, although it contains poorly defined, coarser grained granoblastic patches. Feldspar is interpreted as albite–sericite–carbonate-altered plagioclase, based on faint twinning, and forms tabular phenocrysts up to 2 mm long, some of which are crudely aligned and possibly define a flow foliation. Subhedral biotite is up to 0.4 mm long and forms a decussate aggregate within a 3 mm-wide discontinuous band, which also includes secondary actinolite, pale green chlorite up to 0.5 mm long, and muscovite up to 0.25 mm long. Fine-grained biotite also occurs in linear trails and as disseminated grains throughout the groundmass. Carbonate minerals are anhedral, up to 0.2 mm in size, and form irregular aggregates. Possible ex-ferromagnesian phenocrysts are up to 1 mm in size, and replaced by carbonate and other alteration minerals. Actinolite is colourless to pale green, forms fibrous subhedral laths to 0.2 mm long, some in radiating bunches, and mostly occurs with secondary biotite and carbonate. Pyrite and subordinate pyrrhotite form anhedral irregular masses up to 0.5 mm in diameter and minor chalcopyrite forms disseminated tiny grains and inclusions in pyrrhotite. The sample appears to be a coherent dacitic lava or hypabyssal intrusive rock, and lacks a penetrative tectonic foliation. The abundant secondary biotite and actinolite may reflect contact metamorphism or potassic alteration.

## Zircon morphology

Zircons isolated from this sample are mainly colourless and anhedral to euhedral. The crystals are up to 200 µm long, and equant to slightly elongate, with aspect ratios up to 4:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, and sector zoning is common. A CL image of representative zircons is shown in Figure 2.

## Analytical details

This sample was analysed on 24–25 June 2021, using SHRIMP-B. Fourteen analyses of the M257 standard were obtained during the session, of which 12 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.83% (1σ) and a $^{238}U/^{206}Pb^{*}$ calibration uncertainty of 0.30% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^{*}$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

## Results

Sixteen analyses were obtained from 16 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

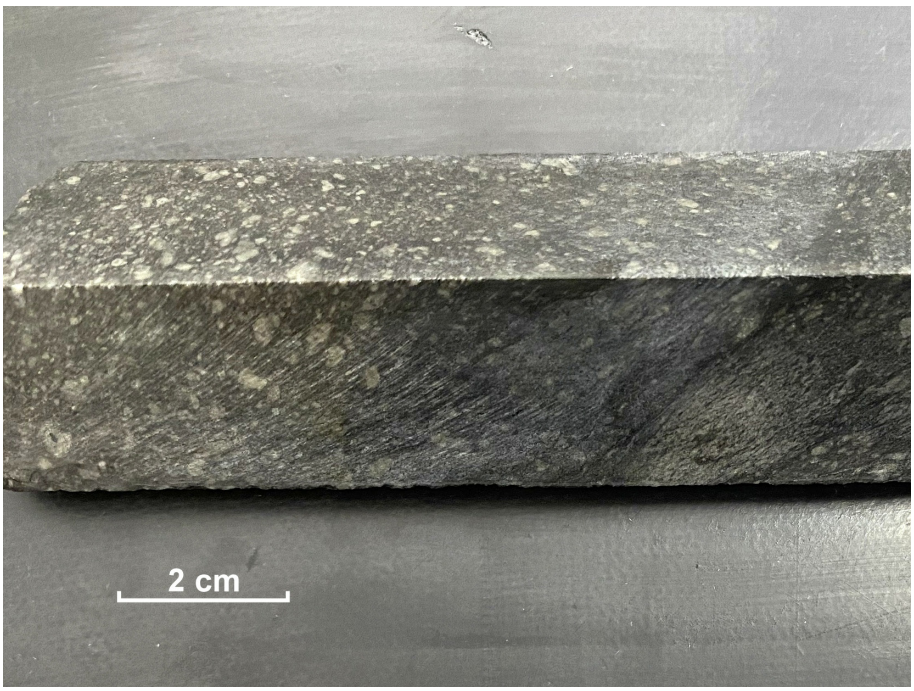


**Figure 1.** **Drillcore image (wet surface) of sample 245127: porphyritic metadacite, Ellen Dam**

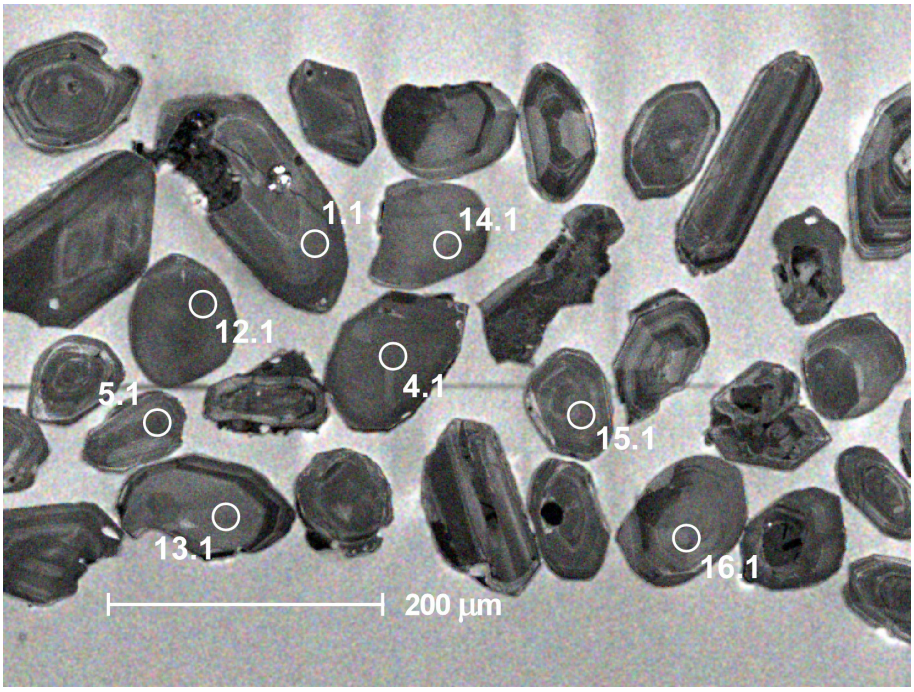


**Figure 2.** **Cathodoluminescence image of representative zircons from sample 245127: porphyritic metadacite, Ellen Dam. Numbered circles indicate the approximate locations of analysis sites**

**Table 1. Ion microprobe analytical results for zircons from sample 245127: porphyritic metadacite, Ellen Dam**

| Group ID | Spot no. | Grain. spot | $^{238}U$ (ppm) | $^{232}Th$ (ppm) | $^{232}Th/^{238}U$ | f204 (%) | $^{238}U/^{206}Pb$ | ± 1σ | $^{207}Pb/^{206}Pb$ | ± 1σ | $^{238}U/^{206}Pb^*$ | ± 1σ | $^{207}Pb^*/^{206}Pb^*$ | ± 1σ | $^{238}U/^{206}Pb^*$ date (Ma) | ±1σ | $^{207}Pb^*/^{206}Pb^*$ date (Ma) | ±1σ | Disc. (%) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| I | 4 | 4.1 | 300 | 487 | 1.62 | 0.082 | 1.943 | 0.021 | 0.18291 | 0.00065 | 1.945 | 0.021 | 0.18218 | 0.00068 | 2675 | 24 | 2673 | 6 | -0.1 |
| I | 1 | 1.1 | 175 | 223 | 1.27 | 0.030 | 2.041 | 0.033 | 0.18256 | 0.00084 | 2.042 | 0.034 | 0.18230 | 0.00085 | 2570 | 35 | 2674 | 8 | 3.9 |
| I | 2 | 2.1 | 388 | 711 | 1.83 | 0.022 | 1.990 | 0.021 | 0.18277 | 0.00057 | 1.991 | 0.021 | 0.18258 | 0.00057 | 2624 | 23 | 2676 | 5 | 2.0 |
| I | 14 | 14.1 | 188 | 242 | 1.29 | -0.011 | 2.015 | 0.024 | 0.18296 | 0.00088 | 2.015 | 0.024 | 0.18305 | 0.00089 | 2598 | 25 | 2681 | 8 | 3.1 |
| I | 9 | 9.1 | 179 | 140 | 0.78 | 0.072 | 1.993 | 0.023 | 0.18371 | 0.00086 | 1.994 | 0.023 | 0.18307 | 0.00090 | 2620 | 26 | 2681 | 8 | 2.3 |
| I | 10 | 10.1 | 211 | 190 | 0.90 | 0.080 | 2.007 | 0.023 | 0.18379 | 0.00080 | 2.009 | 0.023 | 0.18307 | 0.00084 | 2604 | 25 | 2681 | 8 | 2.9 |
| I | 6 | 6.1 | 280 | 428 | 1.53 | 0.021 | 2.015 | 0.022 | 0.18330 | 0.00071 | 2.016 | 0.022 | 0.18312 | 0.00071 | 2597 | 24 | 2681 | 6 | 3.1 |
| I | 11 | 11.1 | 285 | 259 | 0.91 | 0.054 | 2.004 | 0.022 | 0.18371 | 0.00070 | 2.006 | 0.022 | 0.18323 | 0.00072 | 2608 | 24 | 2682 | 7 | 2.8 |
| I | 13 | 13.1 | 152 | 235 | 1.54 | 0.066 | 1.910 | 0.024 | 0.18397 | 0.00098 | 1.912 | 0.024 | 0.18338 | 0.00102 | 2713 | 28 | 2684 | 9 | -1.1 |
| I | 5 | 5.1 | 272 | 271 | 1.00 | 0.000 | 1.971 | 0.021 | 0.18341 | 0.00068 | 1.971 | 0.021 | 0.18341 | 0.00068 | 2645 | 24 | 2684 | 6 | 1.4 |
| I | 8 | 8.1 | 166 | 232 | 1.40 | 0.044 | 2.018 | 0.024 | 0.18389 | 0.00089 | 2.019 | 0.024 | 0.18350 | 0.00092 | 2594 | 26 | 2685 | 8 | 3.4 |
| I | 7 | 7.1 | 282 | 286 | 1.02 | 0.051 | 1.982 | 0.021 | 0.18403 | 0.00069 | 1.983 | 0.021 | 0.18357 | 0.00071 | 2632 | 24 | 2685 | 6 | 2.0 |
| I | 16 | 16.1 | 151 | 199 | 1.32 | 0.013 | 1.933 | 0.024 | 0.18379 | 0.00098 | 1.934 | 0.024 | 0.18367 | 0.00099 | 2687 | 28 | 2686 | 9 | 0.0 |
| I | 12 | 12.1 | 325 | 651 | 2.01 | 0.028 | 2.027 | 0.022 | 0.18405 | 0.00064 | 2.028 | 0.022 | 0.18380 | 0.00065 | 2585 | 23 | 2688 | 6 | 3.8 |
| I | 15 | 15.1 | 310 | 278 | 0.90 | 0.019 | 1.956 | 0.021 | 0.18415 | 0.00068 | 1.957 | 0.021 | 0.18398 | 0.00069 | 2661 | 24 | 2689 | 6 | 1.0 |
| P | 3 | 3.1 | 385 | 670 | 1.74 | 0.163 | 1.952 | 0.020 | 0.18240 | 0.00055 | 1.955 | 0.020 | 0.18095 | 0.00061 | 2663 | 23 | 2662 | 6 | -0.1 |

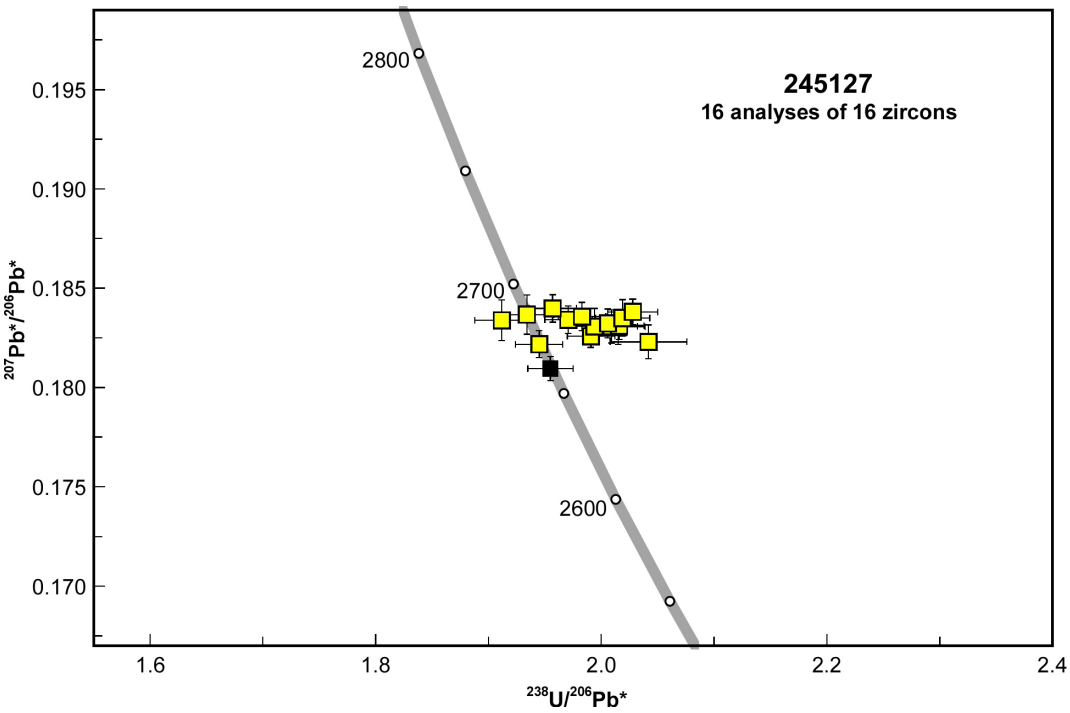


**Figure 3. U–Pb analytical data for sample 245127: porphyritic metadacite, Ellen Dam. Yellow squares indicate Group I (magmatic zircons); black square indicates Group P (radiogenic-Pb loss)**

## Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and can be divided into two groups, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 15 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2682 ± 4 Ma (MSWD = 0.54).

Group P comprises one analysis (Table 1), which yields a $^{207}Pb^*/^{206}Pb^*$ date of 2662 ± 6 Ma (1σ).

The date of 2682 ± 4 Ma for the 15 analyses in Group I is interpreted as the igneous crystallization age of the volcanic rock. The date of 2662 ± 6 Ma (1σ) for the single analysis in Group P is significantly younger and interpreted to reflect loss of radiogenic Pb.

## References

Hall, CE, Wyche, NL and Sapkota, J (partial report), Black Flag Group (A-BF-xf-s): Geological Survey of Western Australia, WA Geology Online, Explanatory Notes extract, viewed 09 February 2021, <www.dmp.wa.gov.au/ens>.

Lu, Y, Wingate, MTD, Sapkota, JK and De Paoli, M 2017, 211051: porphyritic andesite, White Flag Lake; Geochronology Record 1426: Geological Survey of Western Australia, 4p.

Morris, PA 1998, Archaean felsic volcanism in parts of the Eastern Goldfields region, Western Australia: Geological Survey of Western Australia, Report 55, 80p.

Rasmussen, B, Mueller, AG and Fletcher, IR 2009, Zirconolite and xenotime U-Pb age constraints on the emplacement of the Golden Mile Dolerite sill and gold mineralization at the Mt Charlotte mine, Eastern Goldfields Province, Yilgarn Craton, Western Australia: Contributions to Mineralogy and Petrology, v. 157, p. 559–572.

Squire, RJ, Allen, CM, Cas, RAF, Campbell, IH, Blewett, RS and Nemchin, AA 2010, Two cycles of voluminous pyroclastic volcanism and sedimentation related to episodic granite emplacement during the late Archean: Eastern Yilgarn Craton, Western Australia: Precambrian Research, v. 183, no. 2, p. 251–274.

Stacey, JS and Kramers, JD 1975, Approximation of terrestrial lead isotope evolution by a two-stage model: Earth and Planetary Science Letters, v. 26, p. 207–221.

## Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Smithies, RH 2021, 245127: porphyritic metadacite, Ellen Dam; Geochronology Record 1838: Geological Survey of Western Australia, 4p.

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Data released: 10 December 2021